

Searching for Content



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Overview



Finding what you've been looking for:

- Sorting output from **ls**
- Using the **find** command
- Using **locate** to find files
- Searching file content with **grep**
- Using help documentation in Linux



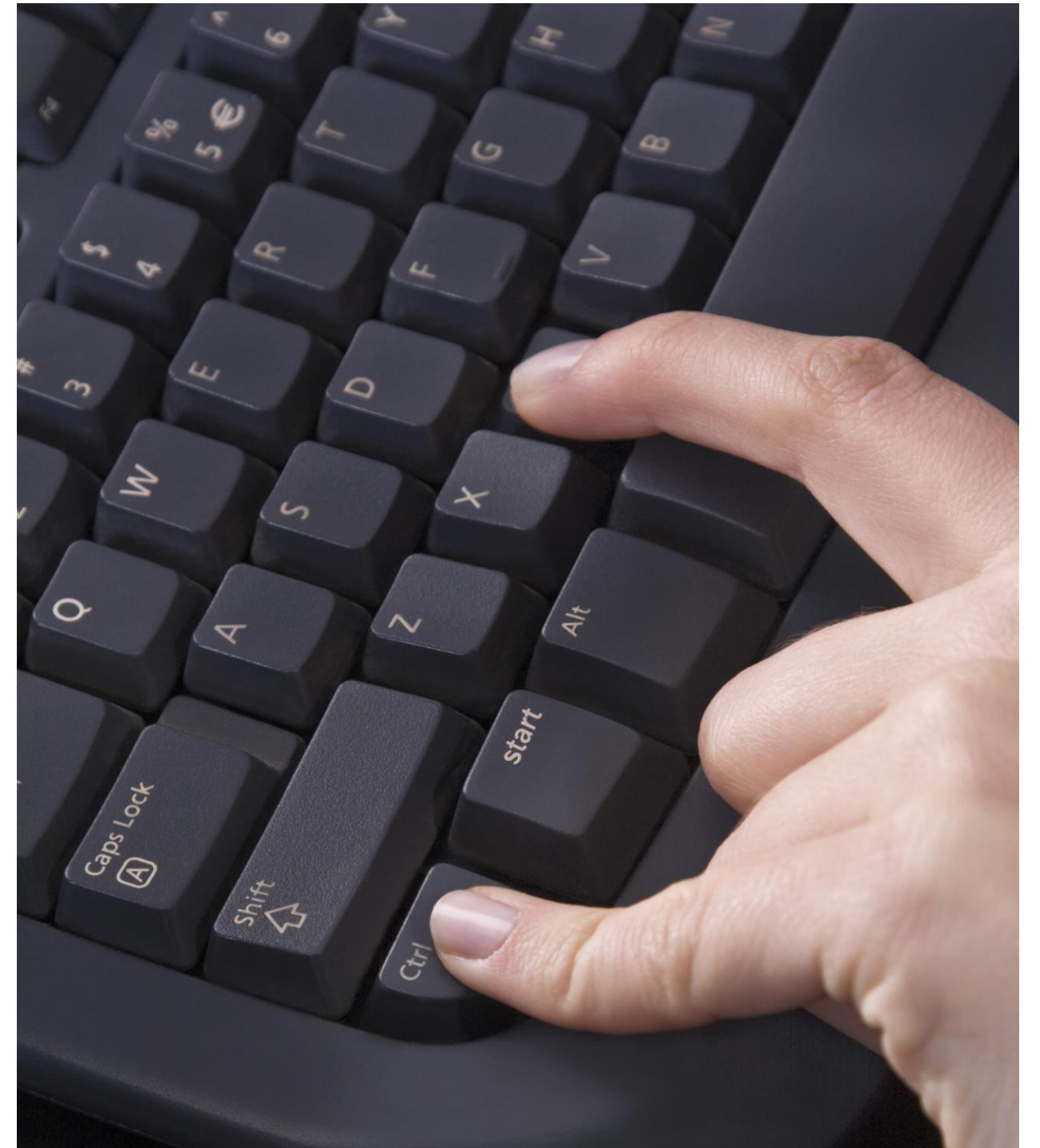
Extra Options



Sorting in Is

Sort most recently edited files: **ls -ltr**

Display largest files last: **ls -lSrh**



Demo



Working with ls:

- Show most recently edited files
- Show largest files



Using Find



```
$ find /etc -mmin 60 2>/dev/null
$ find /etc -mmin -60 2>/dev/null
$ find /etc -mmin +60 2>/dev/null
$ find / -maxdepth 1 -type l
$ find /usr/share/doc -type f -name '*.html' -exec cp {} ~/links/ \;
```

The Power of Find

The find command dynamically searches the given directory structure for files meeting the given criteria. As well as printing we can run commands against each found file.

Demo



Using find at the CLI



Using Locate to Find Files



```
$ sudo apt install -y mlocate  
$ sudo updatedb  
$ locate -S  
$ locate "hosts"  
$ locate -i "hosts"  
$ locate -e -i "hosts"
```

Using Locate

Rather than dynamically searching the complete filesystem we can search a database for efficient operation.

Demo



Using locate



Searching Content



```
$ apt search python3 | wc -l
$ apt search python3 --names-only | wc -l
$ apt search '^python3' --names-only | wc -l
$ apt search '^python3$' --names-only | wc -l
```

Regular Expressions Really Do Help!

As a simple demonstration we can search the package database with apt. We can drill down to just names, excluding the descriptions. The meta character ^ means the line starts with, and \$ means the end lines with.

```
$ grep password /etc/ssh/sshd_config  
$ grep password -i /etc/ssh/sshd_config  
$ grep '^password' -i /etc/ssh/sshd_config  
$ grep '^^[^#].*yes$' -i /etc/ssh/sshd_config
```

Grep Allows Searching of Files

Here are a few examples to entice you into learning more about regular expressions.

Demo



Searching content:

- Regular expressions in apt
- Using grep to search files



Help



```
$ find --help  
$ man find  
$ man 5 passwd
```

Read The Docs

How ever much you may not like this, sometimes the documentation does help!

Demo



Investigating help:

- `--help`
- man pages



Summary



Finding Stuff

- Listing files with ls
- Using find
- Using locate
- Using grep and regular expressions
- Help pages

Up Next:
Archiving Tools and Linux

